

SIXTH SEMESTER DIPLOMA MODEL EXAMINATION IN ENGINEERING/TECHNOLOGY MARCH 2013

WATER QUALITY MANAGEMENT

(Maximum Marks 100)

PART A

Answer the following questions in one or two sentences. Each question carries equal 2marks

1. Define water quality
2. State the tolerance limit of Dissolved Oxygen in drinking water as per IS and WHO standards
3. Name any four microbial organisms polluting water.
4. Mention any four reasons for salinity intrusion
5. What is the function of a coagulant in the process of treating water?

PART B

II. Answer any five of the following. Each question carries 6 marks

1. Identify four physical parameters to be tested for water quality analysis and state their tolerance limit as per ICMR Standards.
2. Briefly mention the requirement of water quality monitoring.
3. Write on 1. Albuminoid Nitrogen 2. Residual chlorine
4. State any six impacts of urbanization on the quality of water.
5. Suggest the different treatments required for the effluent from an oil Refinery.
6. Give an account on the different types of solids and their detection method for water quality analysis.
7. Differentiate between 1. Electrolysis 2. Electro dialysis in terms of waste water treatment.

PART C

(Answer any one full question from each unit. Each question carries 15 marks)

III.

MODULE I

- a) Outline the impact of water quality deterioration on the human health and the environment. (8)
- b) State the tolerance limit of any seven metal minerals in drinking water as per IS standards. (7)

OR

IV .

- a) Print out the different categories of water as per BIS classification and their functional purpose. (7)
- b) Give an account on 1. BIS 2. WHO 3. ICMR standards. (8)

MODULE II

- V a) Give inference on the physical parameters 1. Temperature 2. Taste and odour 3. Colour with reference water quality analysis (9)
- b) State the different steps involved in the water quality monitoring. (6)

OR

VI

- a) Explain the procedure of conducting E coli test in water (9)
- b) What are the main water quality assessment techniques (6)

MODULE III

VII

- a) Give an account on the statement "The industry as a generator of waste" polluting water sources (8)
- b) Define mitigation in terms of salinity intrusion of aquifers. (7)

OR

VIII

- a) Make a relative study on agriculture and water quality of a specific area under consideration (8)
- b) What is Eutrophication ? Suggest the causes for the same (7)

reference water quality analysis

MODULE IV

IX

- a) Write short notes on 1. Membrane filters techniques 2. Water borne diseases (6)
- b) Briefly explain the construction and process of a slow sand filter (9)

OR

X

- a) State the main principles of the following effluent treatment methods (8)
1. Solvent extraction method 2. Adsorption method 3. Chemical precipitation method
- b) Show a schematic diagram for treating the effluents from a oil refinery (7)

